

BAXTER COUNTY AP, AR, USA

WMO: 723439

Lat: 36.369N

Lon: 92.470W

Elev: 283

StdP: 97.97

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-8.1	-18.0	0.8	-8.3	-15.8	1.0	-6.5	8.3	9.5	7.5	9.1	1.9	320	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	36.0	23.6	34.0	23.6	32.6	23.5	25.8	31.7	25.2	31.2	24.6	30.2	2.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.7	28.6	23.5	18.9	28.0	22.8	18.2	27.3	81.0	31.9	78.2	31.1	75.6	30.3	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.9	5.3	-14.8	37.7	2.1	2.6	-16.4	39.5	-17.6	41.0	-18.8	42.5	-20.4	44.3		
			-15.3	26.5	2.2	0.8	-16.8	27.1	-18.1	27.6	-19.3	28.0	-20.9	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	2.9	5.0	9.9	15.2	19.5	24.1	26.2	26.0	21.7	15.6	9.6	4.4
	DBStd	9.46	6.30	6.16	5.73	4.69	3.96	2.86	2.65	2.85	3.84	4.68	5.14	5.54
	HDD10.0	744	232	161	73	11	1	0	0	0	0	8	69	189
	HDD18.3	2091	478	373	268	114	36	1	0	0	14	111	264	433
	CDD10.0	2591	12	22	69	167	296	422	503	496	351	181	57	15
	CDD18.3	896	0	1	5	20	72	174	245	237	114	26	2	0
	CDH23.3	8595	0	4	48	193	574	1618	2507	2371	1037	224	19	1
	CDH26.7	3616	0	0	7	36	148	645	1181	1143	402	52	1	0
Wind	WSAvg	2.1	2.5	2.6	2.8	2.7	2.1	1.7	1.4	1.5	1.5	1.8	2.2	2.3
Precipitation	PrecAvg	1138	83	76	103	120	125	95	93	80	97	85	105	86
	PrecMax	1477	152	156	346	408	255	174	210	175	259	279	317	228
	PrecMin	834	24	17	18	28	35	37	28	7	0	8	14	11
	PrecStd	201	36	37	67	77	51	37	42	45	66	62	78	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.4	23.8	27.4	29.8	32.1	35.2	37.8	38.7	35.3	30.6	26.0	21.2
		MCWB	14.5	14.7	16.3	18.9	21.3	23.8	23.3	22.8	22.3	21.3	17.4	15.3
	2%	DB	18.7	20.5	24.6	27.6	30.0	33.1	35.9	36.1	32.6	28.0	22.7	18.1
		MCWB	13.8	13.3	15.7	17.4	21.6	23.8	23.9	23.5	22.1	19.3	15.4	13.8
	5%	DB	16.0	17.8	22.1	25.8	28.3	32.0	33.9	34.1	31.1	26.1	20.8	15.9
		MCWB	11.2	12.3	14.2	16.7	20.9	23.4	24.1	23.5	21.9	18.3	15.0	11.9
	10%	DB	12.7	15.4	19.5	23.6	26.8	30.4	32.5	32.5	28.9	23.6	18.1	12.9
		MCWB	8.8	10.4	13.1	15.9	20.1	22.9	24.1	23.3	21.0	17.4	13.3	9.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	17.4	18.6	20.5	24.1	26.0	26.7	26.7	24.9	22.8	19.3	17.2
		MCDB	19.8	21.3	25.2	26.7	29.2	32.0	32.3	32.6	30.5	27.9	23.2	19.6
	2%	WB	14.9	15.4	16.9	19.4	22.9	25.1	25.8	25.6	23.9	21.5	17.3	14.9
		MCDB	18.0	18.9	22.5	25.0	27.9	31.0	31.8	31.6	29.1	25.8	20.9	17.8
	5%	WB	11.9	13.3	15.6	18.5	21.9	24.3	25.3	24.9	23.0	20.1	15.8	12.4
		MCDB	14.4	16.7	20.3	23.3	26.7	29.8	31.4	30.8	28.2	23.6	19.5	14.8
	10%	WB	8.9	10.5	14.0	17.3	20.9	23.7	24.7	24.1	22.2	18.7	14.1	10.3
		MCDB	12.0	15.0	18.4	21.7	25.3	28.8	30.5	29.7	27.1	22.4	17.4	12.6
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.9	11.7	12.2	11.1	11.0	11.1	11.7	12.1	12.1	11.1	9.8
		MCDBR	14.6	15.2	15.1	14.8	12.6	12.5	13.3	14.0	14.2	14.3	14.6	12.9
		MCWBR	9.7	9.9	8.2	7.3	6.0	5.2	4.3	4.5	5.3	7.0	8.9	9.0
	5% WB	MCDBR	12.2	12.6	12.4	11.6	10.7	10.8	11.2	11.6	11.4	11.4	12.0	11.4
		MCWBR	9.6	9.7	7.9	6.7	5.7	5.1	4.5	4.5	4.7	6.4	8.6	9.2
Clear-Sky Solar Irradiance	taub	0.306	0.311	0.346	0.381	0.417	0.433	0.463	0.460	0.393	0.349	0.319	0.298	
	taud	2.501	2.479	2.419	2.325	2.284	2.255	2.227	2.207	2.379	2.486	2.521	2.537	
	Ebn at Noon	893	930	921	901	869	852	823	817	860	875	870	881	
	Edh at Noon	81	93	108	126	133	137	140	140	112	92	79	74	
All-Sky Solar Radiation	RadAvg	2.42	3.00	4.07	5.14	5.65	6.46	6.21	5.68	4.86	3.66	2.62	2.06	
	RadStd	0.20	0.37	0.26	0.33	0.50	0.43	0.36	0.37	0.38	0.44	0.29	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	-0.91	+0.55	+1.05	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page